

Summary of game theory so far

- Game theory studies decision making with *strategic interactions*
 - Use *Nash equilibrium* or *stability* to make a prediction
 - Simultaneous-move games: matrix form; dominant/dominated strategy method, best-response method, or guess and verify
 - Sequential-move games: game tree; backward induction
- **Main lessons**
 - Think strategically: put yourself into other peoples' shoes
 - Use a dominant strategy (if any), and never use dominated strategies
 - Individually rational choice can lead to collectively undesirable outcomes
 - Look forward and anticipate your rivals' responses
 - Use credible commitment to change games

PRICE COMPETITION



Yale SCHOOL OF MANAGEMENT

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Price Competition

- Price is an important competition instrument
- We apply game theory to study *strategic pricing decisions* in markets with a few major competitors
 - E.g., airlines, supermarkets, ridesharing
 - Mainly focus on the simple case of *one-shot game*
 - This lays the foundation for more sophisticated price competition you will learn in *Competitor* and *Competitive Strategy*

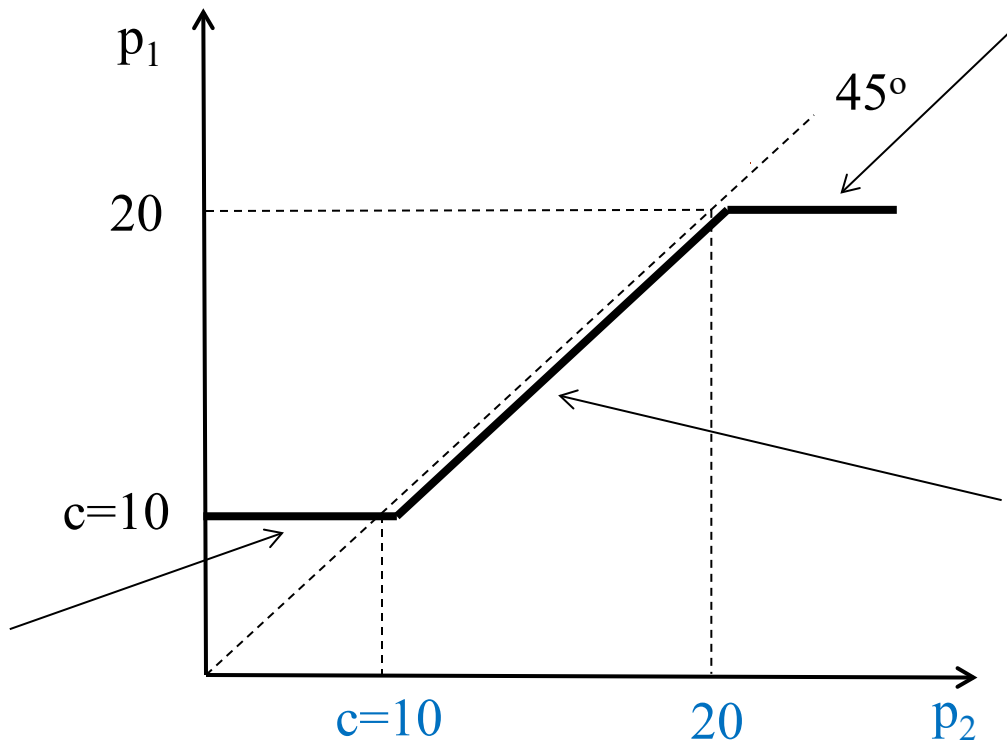
Bertrand Game: Price Competition

[Named after Joseph Bertrand (1822-1900)]

- Two firms 1 and 2 produce an *identical* product
 - The same unit cost, say, **\$10**
 - Set prices p_1 and p_2 simultaneously, *aiming to maximize their own profit*
- Each consumer is willing to pay at most **\$20** for the product
 - The firm asking for a lower price gets the whole market
 - If both offer the same price, they split the market equally
 - Assume 1 million consumers in the market; no capacity constraint
- What price will each firm charge in Nash equilibrium?

Bertrand Game ...

If firm 2 sets $p_2 \leq 10$, firm 1 ensures a zero profit by charging $p_1 = 10$.

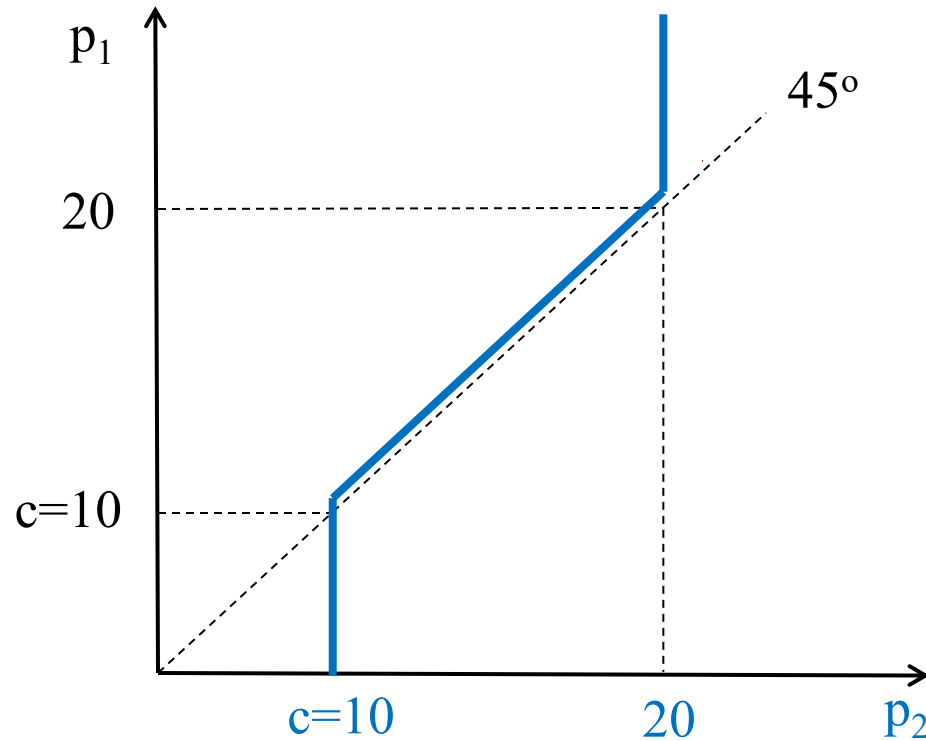


But if firm 2 sets $p_2 > 20$, firm 1 can have the whole market by just charging the monopoly price 20.

If firm 2 sets $p_2 > 10$, firm 1 undercuts slightly to win the whole market.

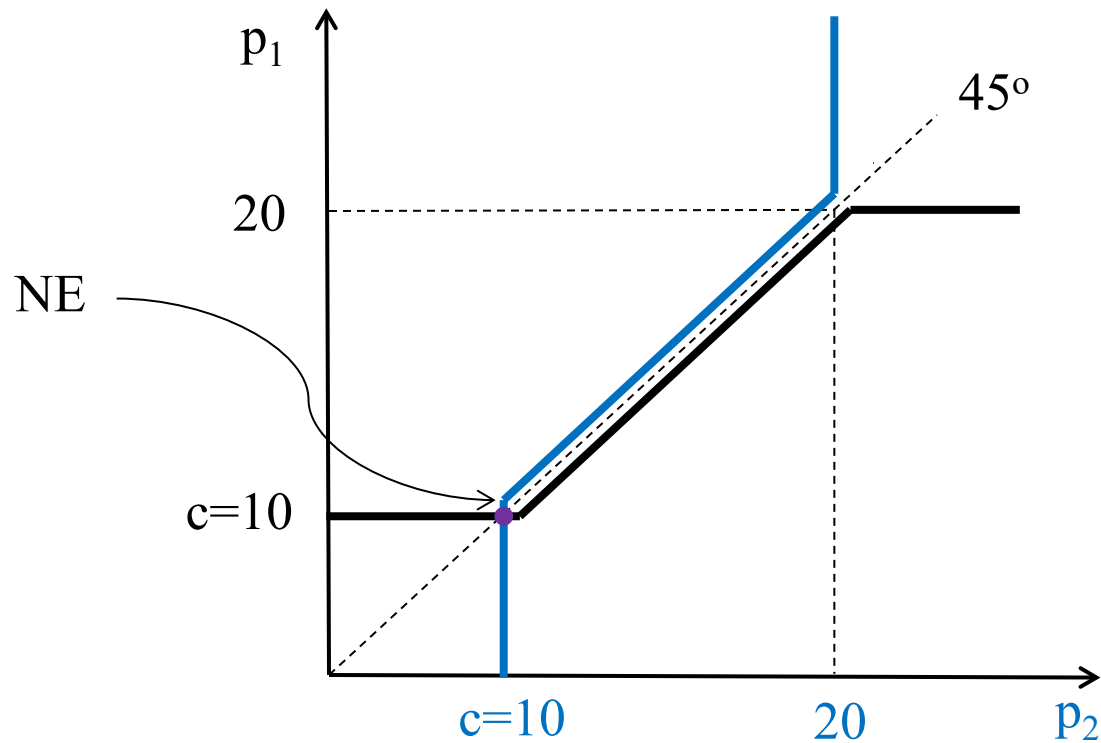
Firm 1's *best-response*: optimal p_1 for each possible p_2

Bertrand Game ...



Firm 2's *best-response*: optimal p_2 for each possible p_1

Bertrand Game ...

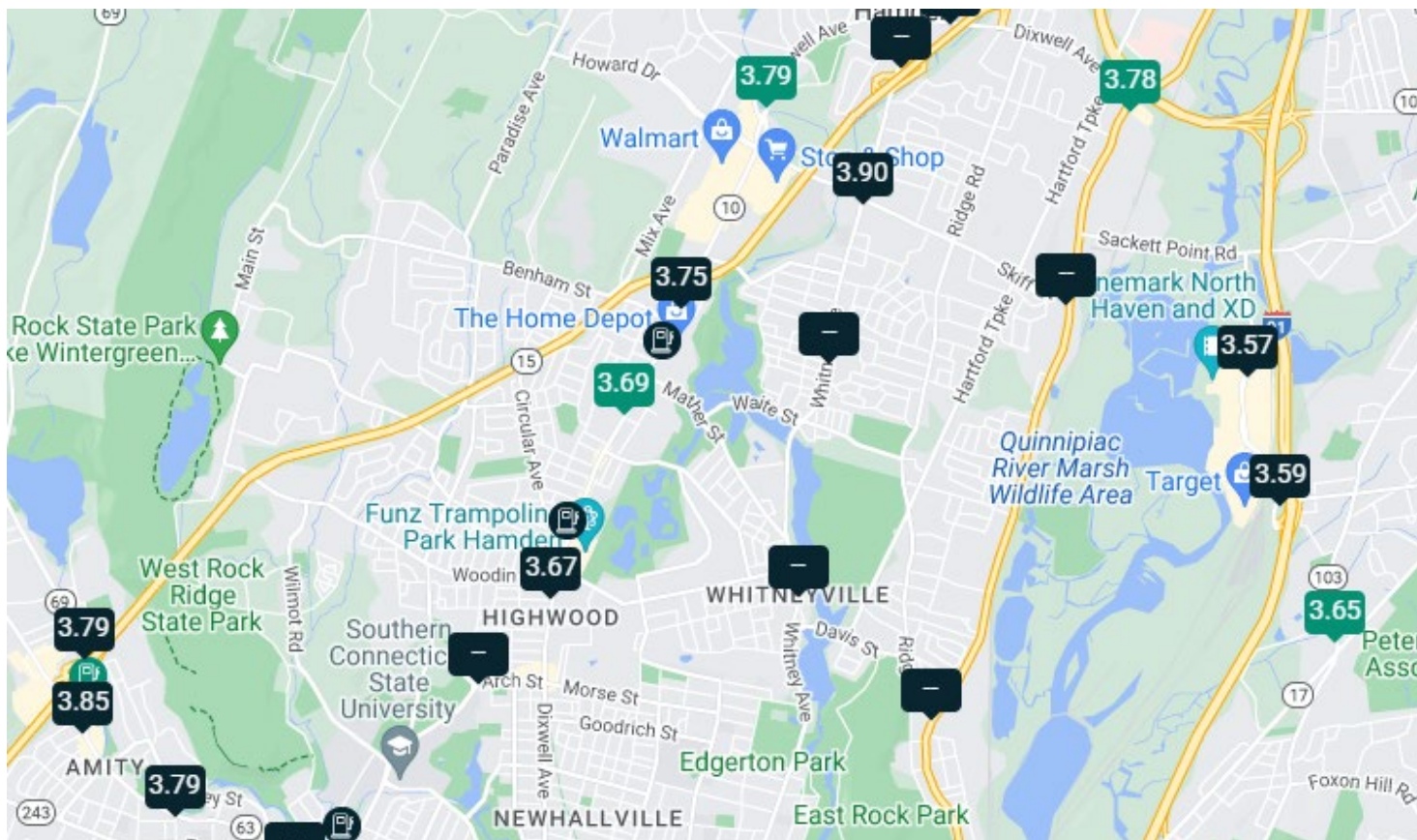


Nash Equilibrium: the intersection of best responses: $p_1 = p_2 = 10$

The “Bertrand Trap”

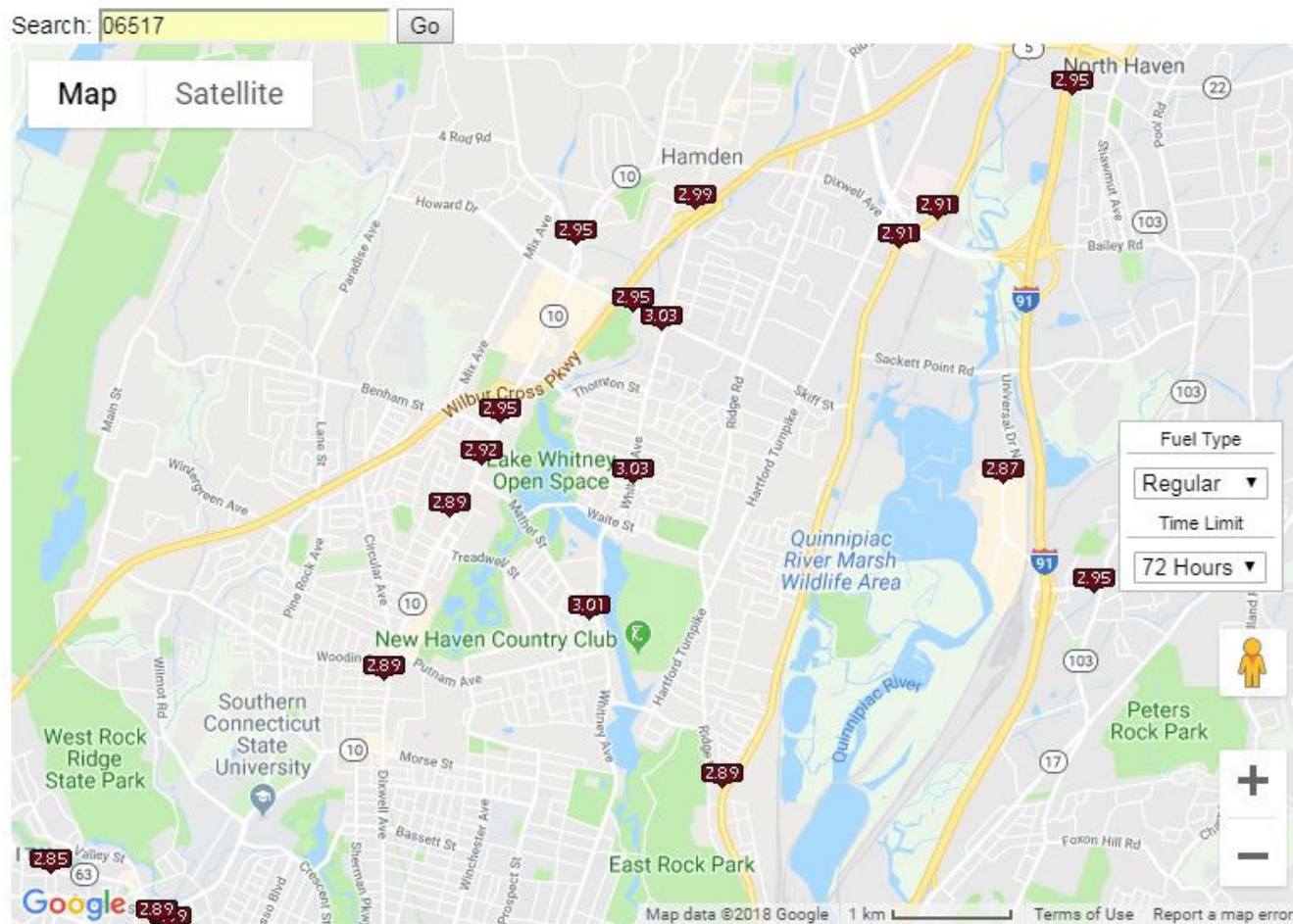
- What’s the NE of this price competition game?
 - Both firms setting prices at the marginal cost \$10
 - If the minimum currency unit is \$0.01, there are two additional NE: (\$10.01, \$10.01) and (\$10.02, \$10.02)
- The danger of the price-cutting game:
 - Even with *only two firms*, price can drop to the level of *competitive price=marginal cost*
 - It is another example of *Prisoner’s dilemma*
- Any real-world examples of Bertrand trap (or price wars)?

Gasoline



ZIP Code: 06517; 8/24/2023

Gasoline



ZIP Code: 06517; 9/13/2018

Trading commission fees

MARKETS

The end of commissions for trading is near as TD Ameritrade cuts to zero, matching Schwab

PUBLISHED WED, OCT 2 2019•8:27 AM EDT | UPDATED WED, OCT 2 2019•4:05 PM EDT

[TD Ameritrade](#) said late Tuesday that the company will eliminate all commission fees for online U.S. stock, exchange-traded fund and option trades. The move came hours after brokerage giant and competitor [Charles Schwab said it is ending commissions](#) for online trading next week. Analysts expect most of the industry to follow suit shortly.

Trading commission fees

E*Trade Joins Rivals in Cutting Commissions to Zero

Beginning Oct. 7, E*Trade will offer zero commissions on stocks, exchange-traded funds and options listed on U.S. exchanges.



E*Trade's competitors have been reducing their fees, in some cases to zero.
PHOTO: CLAUDIO PAPAPIETRO FOR THE WALL STREET JOURNAL

Fidelity Is Latest to Cut Online Trading Commissions to Zero

Wall Street's digitization has reset many of the fundamental costs of investing



Beginning early Thursday, Fidelity stopped charging individual investors commissions on online trades of U.S. stocks, exchange-traded funds and options trades.

PHOTO: BRIAN SNYDER/REUTERS

By [Justin Baer](#)

Updated Oct. 10, 2019 10:24 am ET

- Industry restructuring often follows a price war
 - In 2020, TD Ameritrade acquired by Charles Schwab, and E*Trade by Morgan Stanley

Online Books: Theory Fails?

Price + Shipping	Condition (Learn more)	Delivery	Seller Information
\$104.85 + \$3.99 shipping + \$6.91 estimated tax	New	- Arrives between September 12-18. - Want it delivered Wednesday, September 11? Choose Expedited Shipping at checkout. - Shipping rates and return policy.	TEXTBOOK CHARLIE ★★★★★ 98% positive over the past 12 months. (14,413 total ratings)
\$104.85 + \$5.85 shipping + \$7.03 estimated tax	New	- Arrives between September 12-18. - Want it delivered Wednesday, September 18? Choose Standard Shipping at checkout. - Shipping rates and return policy.	BOOKER C ★★★★★ 100% positive over the past 12 months. (4,734 total ratings)
\$111.89 ✓prime + \$7.11 estimated tax	New	- Free Two-Day Shipping: Get it Sunday, September 8 (order within 18hr 42min) . - Shipping rates and return policy.	amazon.com.
\$119.14 + \$3.99 shipping + \$7.82 estimated tax	New	- Arrives between September 17-24. - Want it delivered Tuesday, September 24? Choose Standard Shipping at checkout. - Shipping rates and return policy.	indoobestsellers ★★★★☆ 87% positive over the past 12 months. (221,874 total ratings)
\$122.45 + \$3.99 shipping + \$8.03 estimated tax	New	- Arrives between September 12-19. - Want it delivered Thursday, September 12? Choose Expedited Shipping at checkout. - Shipping rates and return policy.	Book_Holders ★★★★☆ 84% positive over the past 12 months. (155,640 total ratings)
\$132.17 & FREE Shipping + \$8.39 estimated tax	New	- Arrives between September 13-23. - Want it delivered Monday, September 23? Choose Standard Shipping at checkout. - Shipping rates and return policy.	Book Depository US ★★★★☆ 86% positive over the past 12 months. (860,322 total ratings)
\$148.91 + \$3.99 shipping + \$9.71 estimated tax	New	- Arrives between September 12-19. - Want it delivered Thursday, September 19? Choose Standard Shipping at checkout. - Shipping rates and return policy.	OneStop Deals ★★★★☆ 92% positive over the past 12 months. (2,910 total ratings)
\$234.46 + \$4.99 shipping + \$15.21 estimated tax	New	- Arrives between September 13-20. - Want it delivered Tuesday, September 17? Choose Expedited Shipping at checkout. - Shipping rates and return policy.	Max_Book_Stores ★★★★☆ 87% positive over the past 12 months. (2,536 total ratings)
\$236.25 + \$7.50 shipping + \$15.48 estimated tax	New	- Arrives between Sept. 23 - Oct. 8. - Want it delivered Tuesday, September 17? Choose Expedited Shipping at checkout. - Shipping rates and return policy.	ship_option_#1 ★★★★☆ 80% positive over the past 12 months. (5,920 total ratings)
\$266.08 & FREE Shipping + \$16.90 estimated tax	New	- Arrives between September 13-23. - Want it delivered Thursday, September 19? Choose Standard Shipping at checkout. - Shipping rates and return policy.	booktexts ★★★★★ 98% positive. (3,730 total ratings)

Games of Strategy (Paperback; 9/5/23 on Amazon.com)

Possible Ways Out of the Bertrand Trap

- Make competitors less aggressive:

- Acquisition or merger (may not be permitted by regulator), or cross ownership
- Long-term relationship and collusive pricing (but **illegal** if explicit collusion)
- Be the cost/quality leader
(If $c_1=10$ and $c_2=7$ in the Bertrand game, what is the NE?)
- Limiting capacity
(If there are 3 customers and each firm has capacity 2 in the Bertrand game, what is the NE?)
- Clever pricing strategies (e.g., price matching)
(If both firms offer price matching in the Bertrand game, what is the NE?)

- Make consumers less price sensitive:

- Product differentiation (in many cases even “spurious” differentiation helps)
- Switching costs (e.g., loyalty program)
- Complicating price comparison (e.g., multi-part tariffs)

Availability

Tue, Dec 10

—

Mon, Dec 16

2 adults · 1 child · 1 room

▼

Change search

All available options

Room type	Number of guests	Price for 6 nights	Your choices	Select amount
Basic Two-Bedroom Apartment Recommended for 2 adults, 1 child Select your bed (if available) ☐ 2 twin beds and 1 sofa bed and ☐ 1 queen bed	2 × 5	\$2,632 Excluded: 10.9 % TAX, US\$ 100.00 Resort fee per night	✓ Free cancellation before November 26, 2024 • Pay nothing until November 23, 2024 ✎ Genius discount may be available • Only 7 rooms left on our site	0 ▼

Availability

Tue, Dec 10

—

Mon, Dec 16

2 adults · 1 child · 1 room

▼

Change search

All available apartments

Room type	Number of guests	Price for 6 nights	Your choices	Select amount
All Seasons Two-Bedroom Condo Recommended for 2 adults, 1 child 2 full beds Crib available on request 1250 feet ² Private kitchen Private Bathroom	3	\$2,454 Excluded: 10.8 % Municipality fee, US\$ 69.00 Property service charge per stay, 6.5 % Resort fee	✓ Free cancellation before November 26, 2024 ✎ Genius discount may be available • Only 1 room left on our site	0 ▼

Availability

Tue, Dec 10

—

Mon, Dec 16

2 adults · 1 child · 1 room

▼

Change search

Filter by:

☐ Rooms ☐ Suites

Room type	Number of guests	Price for 6 nights	Your choices	Select amount
Queen Room with Two Queen Beds Recommended for 2 adults, 1 child 2 queen beds 300 feet ² View Air conditioning	3	\$2,498 Excluded: 10.3 % TAX, 0.5 % City tax, US\$ 55.40 Resort fee per night	☑ Breakfast \$39 (optional) ✓ Free cancellation before November 10, 2024 ✓ No prepayment needed – pay at the property • Only 5 rooms left on our site	0 ▼

Availability

Tue, Dec 10

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Mon, Dec 16

2 adults · 1 child · 1 room

▼

Change search

Book this apartment

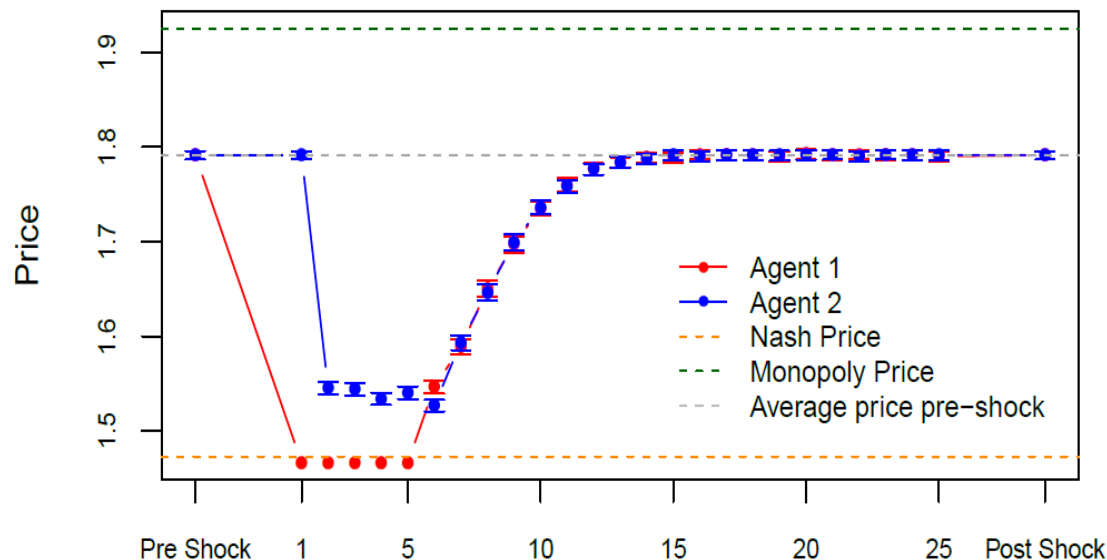
Accommodation Type	Number of guests	Price for 6 nights	Your choices	Select amount
Queen Studio Recommended for 2 adults, 1 child 2 queen beds Private suite 337 feet ² Pool with a view Private Bathroom Coffee machine Sauna Free Wifi	3	\$2,423 Excluded: 9.8 % TAX, US\$ 5.60 Tourism fee per night, 16.62 % Resort fee	✓ Free cancellation before November 10, 2024 • Pay nothing until November 7, 2024 ✎ Genius discount may be available	0 ▼

Confusing hotel fee structures in Vail Co (Booking.com)

Algorithmic pricing and collusion

[Based on Calvano et al. (2020, *American Economic Review*)]

- Lots of debates on whether algorithmic pricing facilitates collusion
- Experimental evidence suggests this tends to be the case



1. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3304991
2. <https://www.wsj.com/articles/beware-algorithms-that-could-collude-on-prices-11554155892>
3. <https://www.wsj.com/articles/why-do-gas-station-prices-constantly-change-blame-the-algorithm-1494262674>

Price Competition Summary

- Price cutting is a dangerous game (but good for consumers)
 - Another example of [Prisoner's Dilemma](#)
- Finding ([legal](#)) ways to escape from Bertrand trap is important for business
- Repeated interactions in a long-term relationship may help firms “cooperate” with each other
- Game theory can also help study competition in other dimensions such as quantity, quality, advertising, and R&D

Materials for exam preparation:

- Slides and class notes
- Q2 in PS3, and the extra problem set
- Two past exam papers

- Please email whenever you have
 - questions about game theory
 - good examples or articles to share

Thank you!

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Enjoy the rest of you journey at SOM!